Real Estate Support Triage Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

Real estate companies frequently receive a large number of customer inquiries related to property purchases, rental availability, property visits, maintenance complaints, and general information. Managing these inquiries manually can be time-consuming and inefficient, often leading to delayed responses, misclassification of requests, and poor customer experience.

Traditional customer support systems rely heavily on human operators to read, interpret, and route each inquiry to the appropriate department. As the volume of inquiries increases, it becomes difficult to maintain quick response times and ensure that urgent issues are prioritized effectively.

To address this challenge, there is a need for an intelligent system that can automatically analyze customer inquiries, identify their intent and urgency, extract relevant details such as property references or appointment dates, and generate appropriate responses. The **Real Estate Support Triage Agent** aims to solve this problem by using an AI-powered multi-agent pipeline to automate the triage process and assist real estate support teams in handling inquiries efficiently

1.2 Project Objectives

The main objectives of the Real Estate Support Triage Agent are:

1. **Automate Inquiry Classification**
   Automatically analyze incoming customer inquiries and determine their intent and urgency using an AI-based multi-agent system.
2. **Extract Structured Information**
   Identify and extract important entities such as property IDs, dates, or references from unstructured natural language queries.
3. **Generate Draft Responses**
   Produce AI-generated professional responses that can assist customer support teams in replying quickly to user inquiries.
4. **Improve Support Efficiency**
   Reduce the manual workload of real estate support staff by automating the initial triage of inquiries.
5. **Provide Inquiry Tracking**
   Allow users to track the status of their submitted inquiries and maintain a history of requests.

6. **Enable Administrative Monitoring**
   Provide an admin dashboard where administrators can monitor, prioritize, and update inquiry statuses.

1.3 Scope of the Project

The scope of the Real Estate Support Triage Agent includes the design and development of a full-stack web application capable of automating the processing of real estate customer inquiries.

The system allows users to register, log in, and submit property-related inquiries through a web interface. These inquiries are processed by an AI-powered triage pipeline that classifies the request, extracts relevant information, and generates a draft response. The processed data is stored in a database for tracking and administrative review.

The platform also provides an administrative dashboard that enables administrators to monitor incoming inquiries, view AI-generated analysis, and update the status of requests based on support actions.

The system focuses on improving the **initial triage and management of real estate inquiries**, while human support teams remain responsible for final decision-making and resolution of complex issues.

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2. Proposed Solution

2.1 Key features

The Real Estate Support Triage Agent provides several intelligent features designed to automate and streamline the handling of customer inquiries in real estate support systems. The key features of the system include:

- **AI-Based Inquiry Classification**
  Automatically classifies user inquiries based on intent (e.g., buying, viewing, complaint, general inquiry) and urgency level.
- **Multi-Agent AI Processing Pipeline**
  Uses a multi-agent architecture where specialized AI agents collaborate to analyze customer inquiries and generate structured outputs.
- **Information Extraction from Queries**
  Extracts important entities such as property IDs, appointment dates, and references from natural language input.

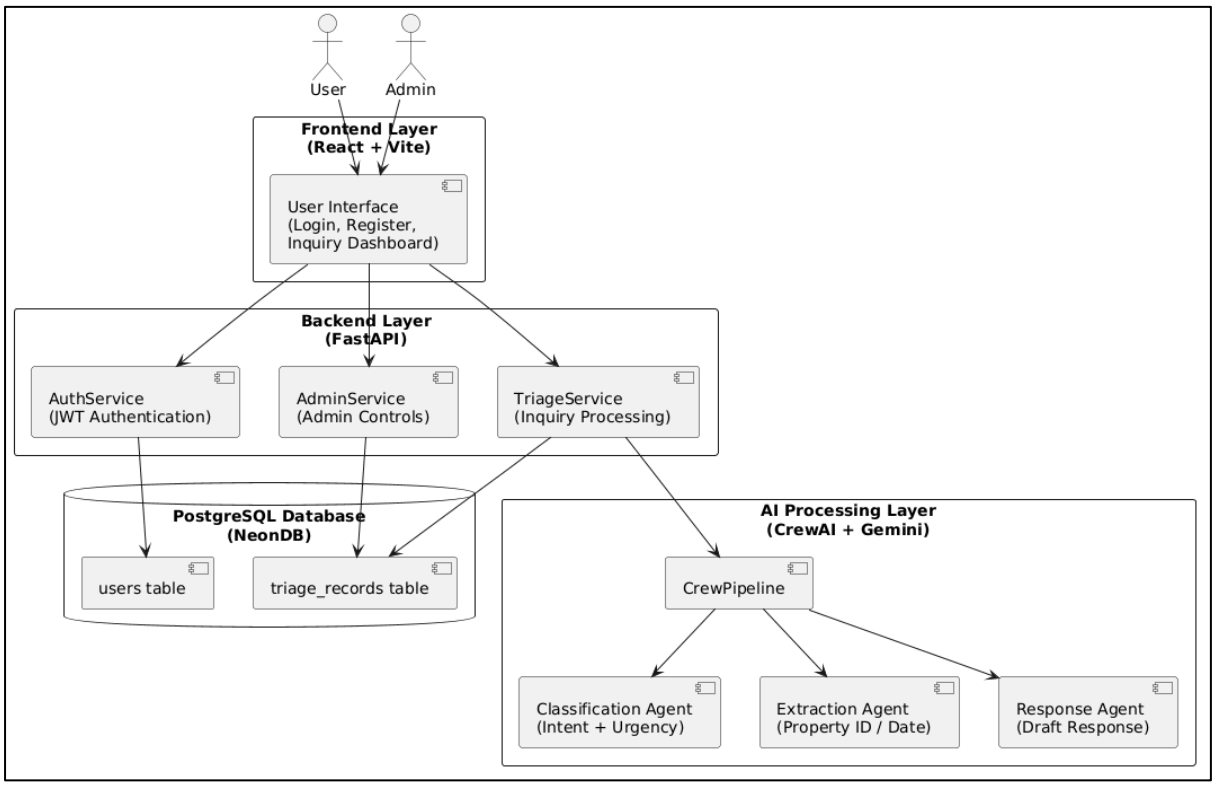
- **AI-Generated Draft Responses**
  Generates professional draft responses to assist support teams in responding quickly to customer inquiries.
- **User Authentication System**
  Provides secure user registration and login functionality using JWT-based authentication.
- **Inquiry Submission Dashboard**
  Allows users to submit real estate inquiries and receive automated analysis of their requests.
- **Inquiry History Tracking**
  Enables users to view their previously submitted inquiries and monitor their status.
- **Administrative Monitoring Dashboard**
  Provides administrators with a dashboard to view, monitor, and manage customer inquiries.
- **Inquiry Status Management**
  Allows administrators to update the status of inquiries (e.g., Unsolved, Action Taken, Solved).
- **Layered and Modular Architecture**
  The system follows a layered architecture with separate frontend, backend, AI processing, and database layers to ensure scalability and maintainability.

2.2 Overall Architecture / Workflow

The Real Estate Support Triage Agent follows a layered full-stack architecture consisting of a frontend interface, backend services, an AI processing layer, and a database layer. The frontend is developed using React with Vite and provides interfaces for user registration, login, inquiry submission, and inquiry tracking. Users interact with the system through this interface to submit real estate–related inquiries such as property visits, complaints, or information requests. These requests are sent to the backend through RESTful APIs.

The backend is implemented using FastAPI and acts as the central application layer that manages authentication, business logic, and communication with the AI processing pipeline. When a user submits an inquiry, the backend verifies the user's identity using JWT authentication and forwards the inquiry to the AI triage pipeline. The pipeline uses a multi-agent architecture implemented with CrewAI and powered by the Gemini language model. The agents work sequentially to classify the intent and urgency of the inquiry, extract relevant information such as property references or appointment dates, and generate a professional draft response.

After processing, the structured results are returned to the backend and stored in a PostgreSQL database along with the original inquiry and generated response. The processed response is then displayed to the user through the frontend interface. Additionally, administrators can access a dedicated dashboard where they can monitor incoming inquiries, review AI-generated analysis, and update the status of requests. This architecture ensures efficient processing of customer inquiries while maintaining a clear separation between presentation, application logic, AI processing, and data storage layers.



2.3 Tools & Technologies Used

The Real Estate Support Triage Agent is developed using a combination of modern web technologies, AI frameworks, and database systems to ensure efficient processing of customer inquiries. The system follows a full-stack architecture where different technologies are used for the frontend interface, backend services, AI processing pipeline, and data storage.

The frontend of the application is developed using **React with Vite**, which provides a fast and responsive user interface for user authentication, inquiry submission, and request tracking. The backend services are implemented using **FastAPI**, a high-performance Python framework that handles API requests, authentication, and communication with the AI pipeline.

For AI-based inquiry analysis, the system uses **CrewAI**, which enables the creation of a multi-agent pipeline where different agents collaborate to process user inquiries. The agents are powered by the **Google Gemini language model**, which performs natural language understanding tasks such as intent classification, information extraction, and response generation.

The system uses **PostgreSQL** as the primary database for storing user accounts and inquiry records. Database schema management and version control are handled using **Alembic migrations**, while **SQLAlchemy ORM** is used for interacting with the database through Python objects instead of raw SQL queries. Additionally, **JWT (JSON Web Tokens)** are used to implement secure authentication and stateless session management within the system.

Category	Technology / Tool	Purpose
Frontend Framework	React (with Vite)	Builds the user interface for registration, login, and inquiry submission
Backend Framework	FastAPI	Handles REST APIs, authentication, and business logic
Programming Language	Python	Used to implement backend services and AI processing
AI Framework	CrewAI	Manages the multi-agent AI pipeline for inquiry triage
Large Language Model	Google Gemini	Performs intent classification, entity extraction, and response generation
Database	PostgreSQL (NeonDB)	Stores user accounts and inquiry records
ORM	SQLAlchemy	Enables interaction with the database using Python models
Database Migration	Alembic	Manages database schema versioning and updates
Authentication	JWT (JSON Web Token)	Provides secure user authentication and session management
API Communication	REST APIs	Enables communication between frontend and backend services

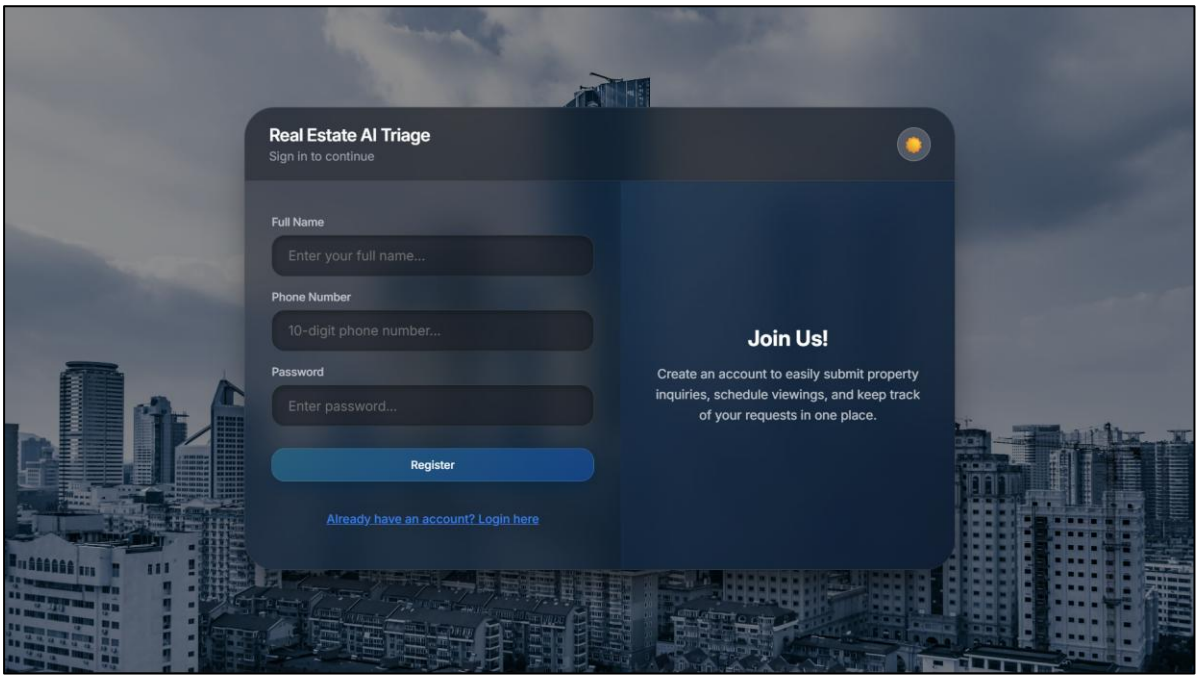
3. Results & Output

This section presents the outputs and results obtained from the implementation of the **Real Estate Support Triage Agent**. The system successfully demonstrates how AI-powered automation can assist real estate companies in efficiently managing and responding to customer inquiries.

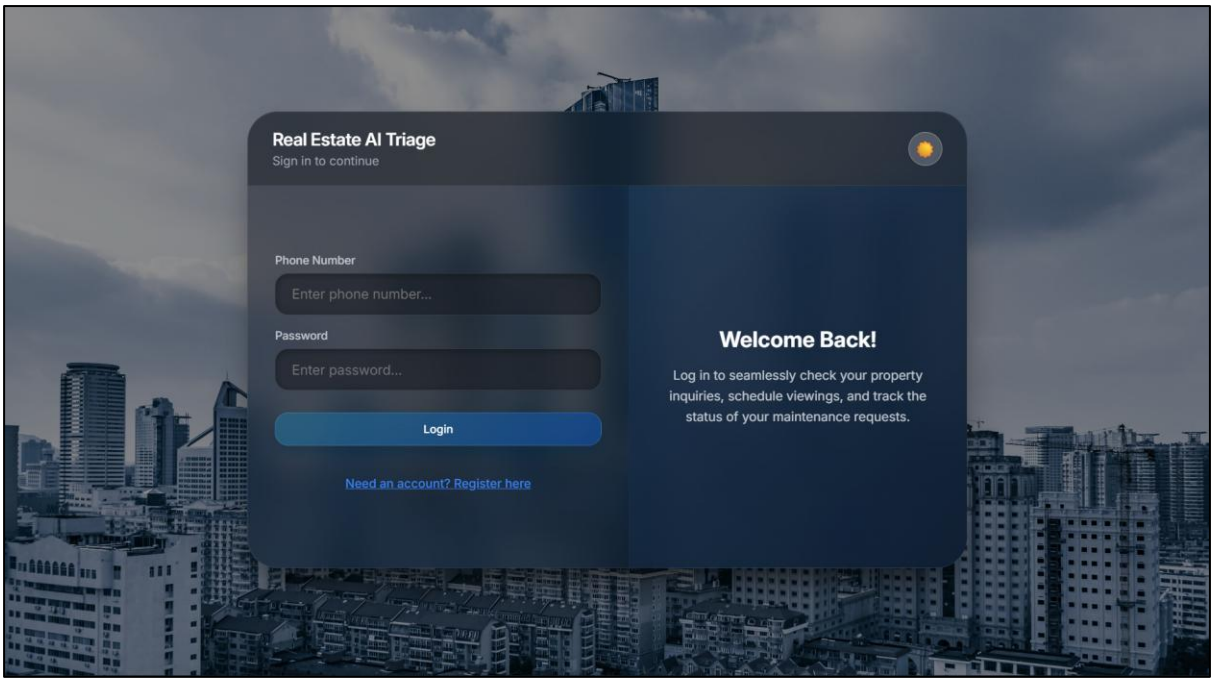
3.1 Screenshots / Outputs

The developed system includes several functional interfaces that demonstrate the working of the platform:

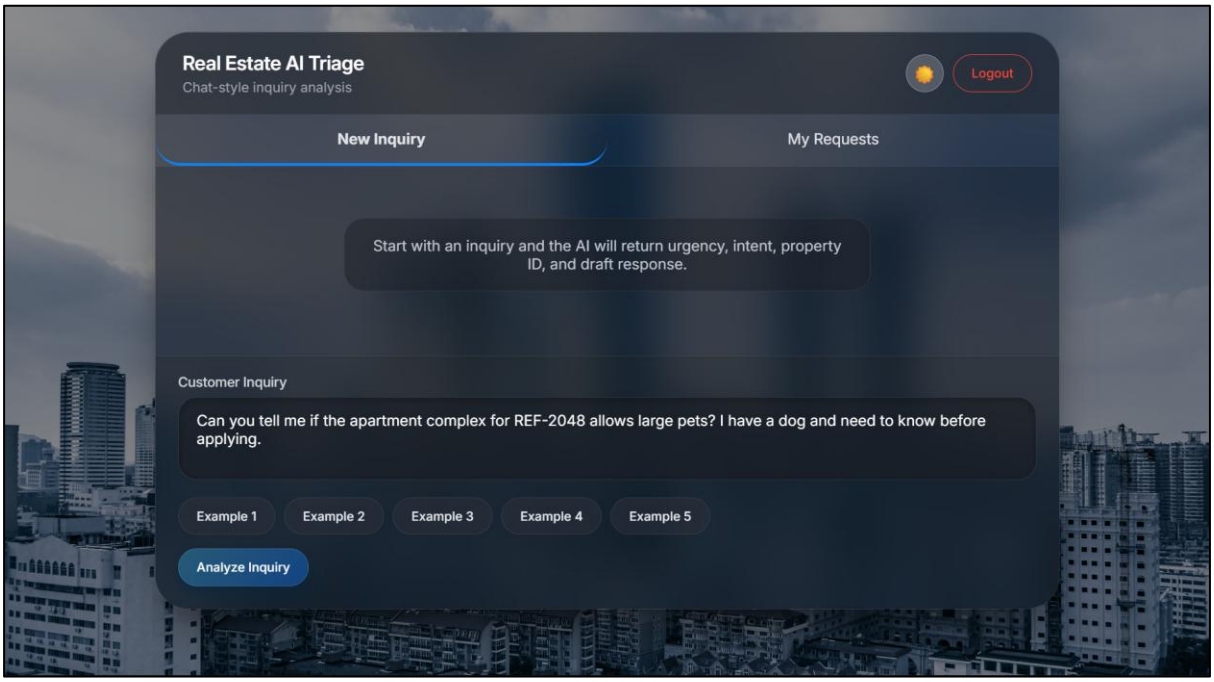
- **User Registration Interface**
  Allows new users to create an account by providing their name, phone number, and password.



- **User Login Interface**
  Enables registered users to securely authenticate and access the system using JWT-based authentication.

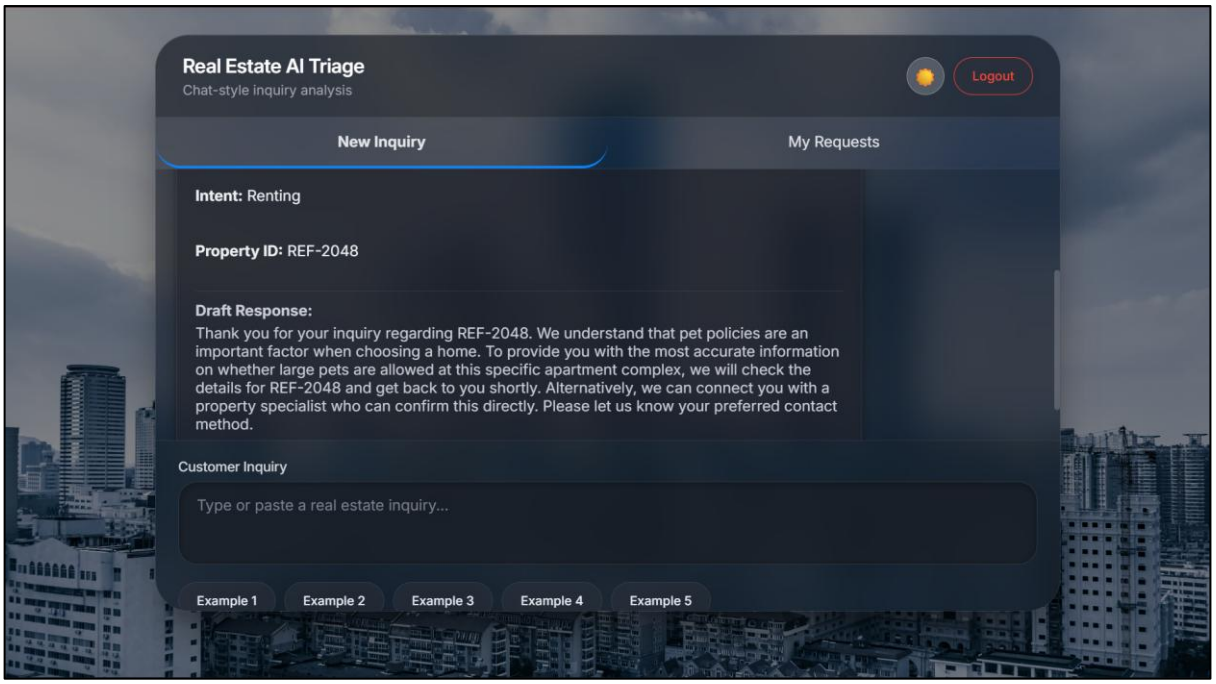


- **Inquiry Submission Dashboard**
  Users can submit real estate inquiries through an interactive interface. Once submitted, the inquiry is processed by the AI triage pipeline.

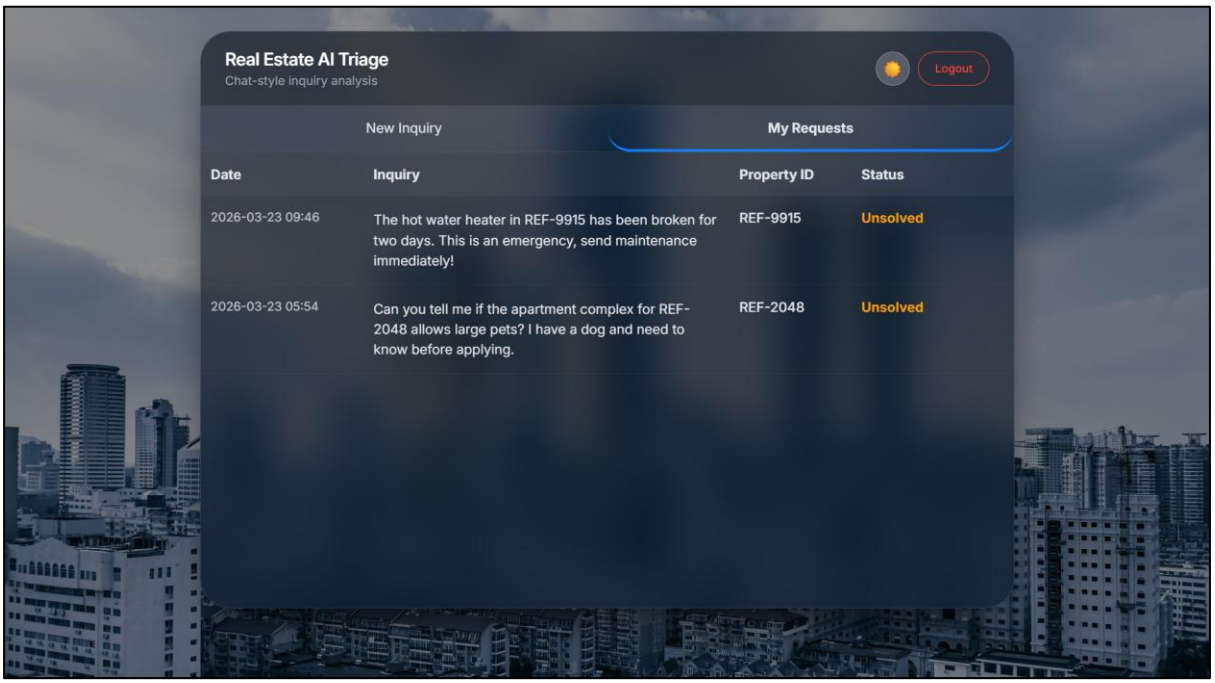


- **AI Analysis Output**
  After processing the inquiry, the system displays structured results including:
  - Urgency level of the inquiry
  - Classified intent (e.g., buying, viewing, complaint)

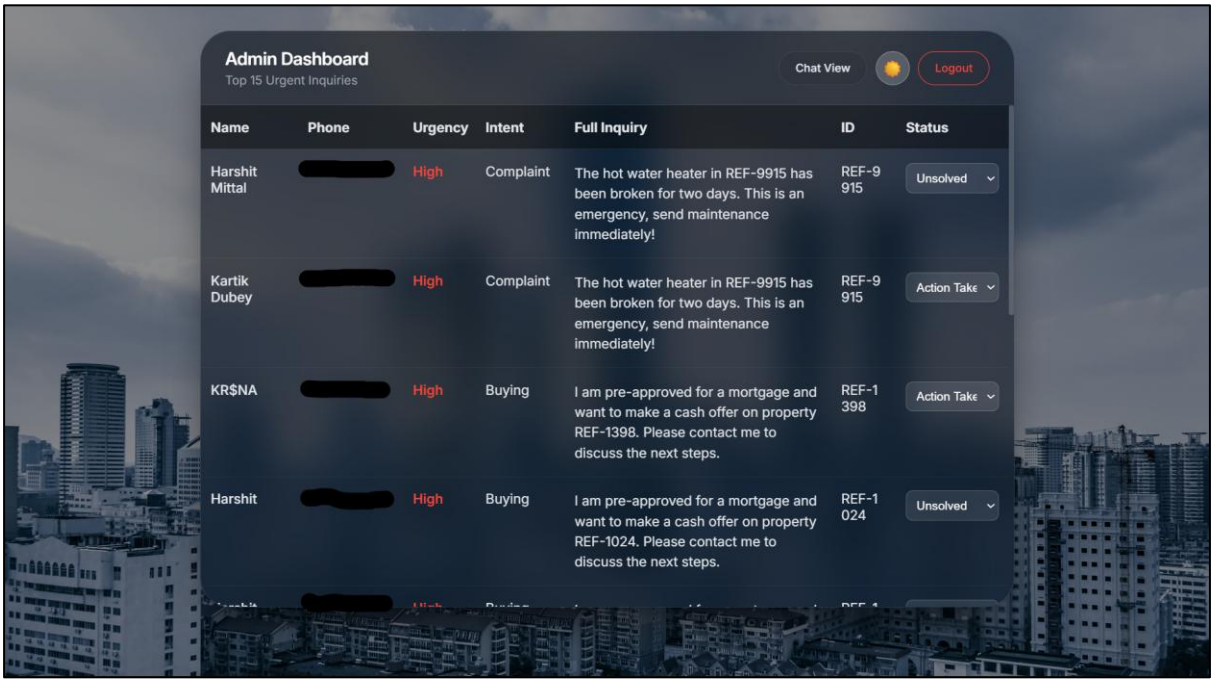
- Extracted property reference or appointment date
- AI-generated draft response



- **User Request History Page**
  Users can view previously submitted inquiries along with their current status (Unsolved, Action Taken, or Solved).



- **Administrative Dashboard**
  Administrators can view and manage all incoming inquiries, including details such as user information, inquiry text, urgency classification, and generated responses.



3.2 Dashboards

The system includes an **administrative monitoring dashboard** that allows administrators to monitor customer inquiries and manage their resolution process.

Key functionalities of the dashboard include:

- Viewing a list of all customer inquiries submitted to the system
- Displaying user details such as name and phone number
- Showing AI-classified urgency levels and inquiry intent
- Viewing extracted property references and AI-generated responses
- Updating the status of inquiries to track support progress

The AI processing pipeline is built using a **multi-agent architecture**, where specialized agents collaborate to analyze inquiries. The system includes the following agents:

- **Classification Agent** – Determines the intent and urgency of the inquiry.
- **Extraction Agent** – Extracts structured information such as property IDs and dates.
- **Response Agent** – Generates a professional draft response for the user.

This agent-based architecture enables efficient processing of customer queries while maintaining structured outputs for administrative tracking.

3.3 Key Outcomes

The implementation of the Real Estate Support Triage Agent produced several key outcomes:

- Successful development of a **full-stack AI-powered inquiry management system**.
- Automated classification and analysis of customer inquiries using a **multi-agent AI pipeline**.
- Efficient extraction of relevant information from natural language queries.
- Generation of AI-assisted responses to support customer service teams.
- Secure authentication and role-based access control for users and administrators.
- Administrative tools for monitoring and prioritizing customer inquiries.

Overall, the system demonstrates how AI-based automation can significantly improve the efficiency of customer support operations in the real estate domain by reducing manual workload and improving response management.

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4. Conclusion

The **Real Estate Support Triage Agent** successfully demonstrates the use of artificial intelligence to automate and improve the handling of customer inquiries in the real estate domain. The system integrates a full-stack web architecture with an AI-powered multi-agent pipeline to analyze user queries, classify their intent and urgency, extract relevant information, and generate professional draft responses.

By combining modern technologies such as React for the frontend, FastAPI for backend services, and a CrewAI-based multi-agent system powered by the Gemini language model, the platform efficiently processes customer inquiries and organizes them in a structured manner. The system also provides secure authentication mechanisms, inquiry tracking for users, and an administrative dashboard that allows support teams to monitor and manage incoming requests effectively.

The implementation of this system demonstrates how AI-driven automation can significantly reduce manual effort in customer support operations while improving response efficiency and consistency. The modular and scalable architecture of the system also allows for future enhancements and integration with other enterprise systems used in the real estate industry.

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5. Future Scope & Enhancements

Although the Real Estate Support Triage Agent successfully automates the initial processing of customer inquiries, several enhancements can be implemented in future versions to further improve the system's capabilities, scalability, and usability.

One possible improvement is the integration of **advanced AI models and additional agents** to perform deeper analysis of user inquiries. Additional agents could be introduced to detect sentiment, predict customer urgency more accurately, or recommend personalized property suggestions based on user preferences.

The system can also be enhanced by integrating **notification mechanisms** such as email or SMS alerts to inform users when the status of their inquiries changes or when an administrator responds to their request. This would improve communication between customers and real estate support teams.

Another potential enhancement is the development of **analytics and reporting dashboards** for administrators. These dashboards could provide insights into inquiry trends, frequently asked questions, response times, and customer satisfaction metrics, helping organizations improve their customer support strategies.

Future versions of the system could also integrate with **external CRM systems used by real estate companies**, allowing inquiries processed by the AI triage system to be automatically synchronized with existing customer relationship management platforms.

Additionally, **caching mechanisms and distributed deployment strategies** could be introduced to improve performance and scalability when the system handles a large volume of customer inquiries. Technologies such as Redis or distributed microservices architecture could support this scalability.

Overall, these enhancements would further strengthen the system's ability to provide intelligent, scalable, and efficient support automation for real estate organizations.